Lab 2 b

Depth - First Search (DFS)

Imagine the same group of children is participating in a treasure hunt inside a large school building. The classrooms in the school are named A, B, C, D, E, F, G and H. The children begin their search from classroom A, and their goal is to find the treasure hidden in classroom H. Everyroom is connected to only a few other classrooms through doors. The children know which classrooms are connected, but they do not know the exact path that leads to the treasure. Therefore, they decide to follow a fixed searching method so that they do not get confused or keep waiting in circles.

This time, the children decide that whenever they enter a classroom, they will continue along that path as long as there is another class infront of them. They do not stop to check other nearby classrooms imediately. Their idea is very simple: keep moving forward until there is nowhere else to go, and only then return to try another path. This is different from the previous method, where every nearby classroom was explored before moving farther away.

The children start their journey from classroom A. From there, they saw two possible classrooms, B and C. Instead of checking both of them, they choose B and continue moving forward. When they reach B, they again find two more classrooms, D and E. Following the same idea, they continue with D first because it is the next available classroom on that path.

After entering classroom D, they look around and realize that there are no more doors leading to any other classroom. Since they have reached the end of that path, they cannot continue any further. They now return to the previous classroom, which is B, and continue with the remaining unexplored classroom, E.

When they enter classroom E, they notice another door leading to H. They enter classroom H and finally discover the treasure. The path they followed to reach the treasure is A→B→E→H. Even though, they successfully found the treasure, they never visited classrooms C, F and G because they found the goal before needing to explore that side of the building.

This searching method is very similar to how a person explores a maze. Imagine walking through a maze where, every time you see a new passage, you keep walking through it without turning back. Only after reaching a dead end, you return to the previous junction and try another passage. By following this method, you complete one path entirely before beginning another. This is the basic idea behind Depth-First Search.

Now let us understand how this idea is represented in the program. The classrooms and their connections are stored in a graph. Each classroom is connected to the classrooms that can be reached directly from it. The search begins at classroom A, and the goal is to reach classroom H.

The program keeps a list called stack, which stores all the possible paths that can still be explored. At the beginning, this stack contains only the path ['A'], because the search starts from classroom A. Unlike BFS, the program always chooses the last path that was added to this list.

path = stack.pop()

The pop() function removes the last element from the stack. This follows the LIFO principle. It is similar to stack of books. Because of this rule, the search immediately continues with the newest path, allowing it to move deeper and deeper before returning.

For this graph, the program returns the path:

['A', 'B', 'E', 'H']

The order of visiting the classrooms is:

['A', 'B', 'D', 'E']

Notice that the search reaches the goal without visiting C, F and G because the treasure is found before those classrooms need to be explored.